

1. Model Selection & Justification

Four models were selected to cover a spectrum of complexity, supporting the SDG 4 objective of predicting dropout:

- **Logistic Regression** — Linear baseline; transparent coefficients easily explainable to academic advisors.
- **Random Forest** — Robust ensemble handle non-linear interactions; provides reliable feature importance scores.
- **Gradient Boosting** — Boosting of weak learners; captures subtle patterns in semester-by-semester performance.
- **Neural Network (MLP)** — Two-hidden-layer network serving as a deep-learning representation benchmark.

2. Data Preparation for Modelling

The UCI dataset (4,424 students, 36 features, 0 missing values) was prepared as follows:

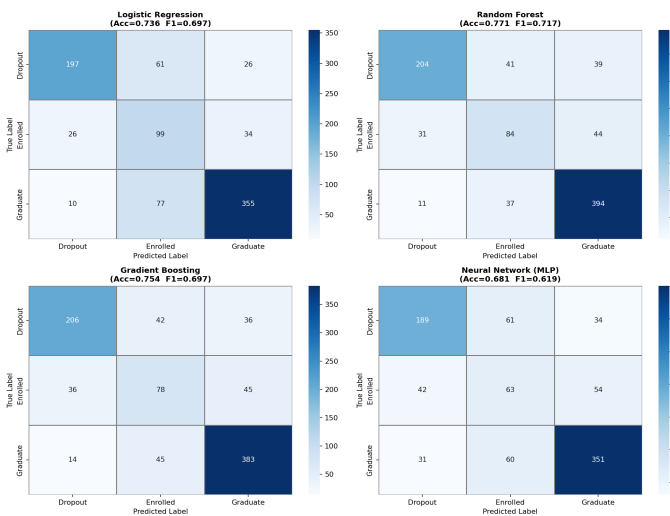
- **Train/Test Split:** 80% training / 20% test, stratified to preserve the 49.9% Graduate / 32.1% Dropout / 17.9% Enrolled ratio.
- **Feature Engineering & Normalisation:** All features retained; Target label-encoded; StandardScaler applied to metrics.
- **Class Imbalance:** SMOTE applied to training set only, balancing all classes to equal frequency without test leak.

3. Model Training & Evaluation

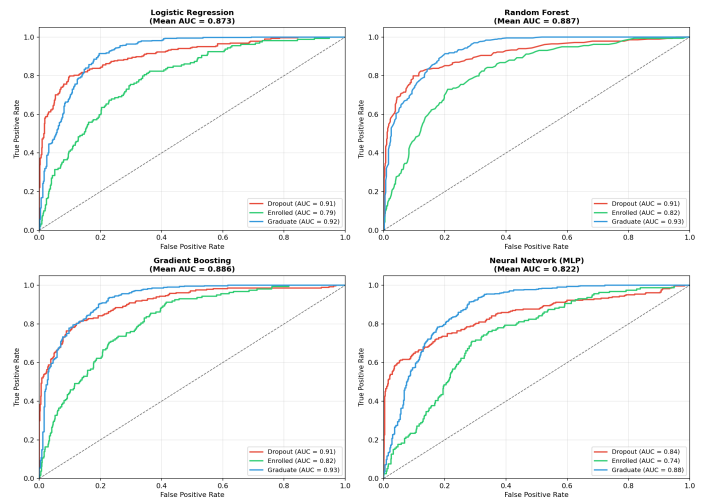
Models were trained on the SMOTE-balanced training set and evaluated on the held-out test set using 5-fold CV:

Model	Accuracy	Precision	Recall	F1-Score (Macro)	CV F1 ± Std
Logistic Regression	0.7356	0.7062	0.7065	0.6969	0.7425 ± 0.0209
Random Forest	0.7706	0.7246	0.7127	0.7169	0.8417 ± 0.0168
Gradient Boosting	0.7537	0.7009	0.6941	0.6966	0.8296 ± 0.0121
Neural Network (MLP)	0.6814	0.6211	0.6186	0.6188	0.8557 ± 0.0167

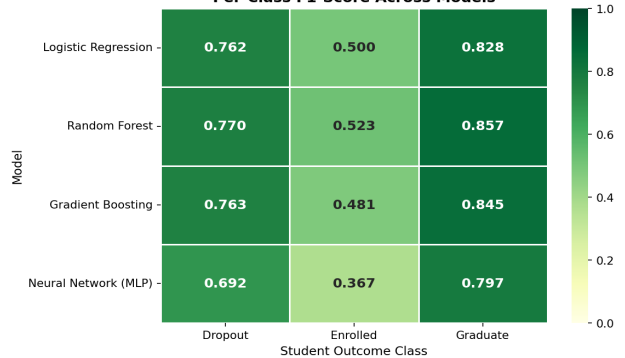
Confusion Matrices — All Models



ROC Curves (One-vs-Rest) — All Models



Per-Class F1-Score Across Models



4. Interpretation, Limitations & Improvements

Key findings: Random Forest achieved the best test performance (Acc=77.1%, F1=0.717). The 'Enrolled' class proved hardest to classify across all models (F1=0.37–0.52), likely because enrolled students are mid-journey. ROC confirms strong discriminability for Dropout vs. Graduate (AUC>0.88), which is operationally crucial. Top predictors are 2nd-semester performance, tuition status, and scholarship holding.

Limitations & Improvements: MLP showed signs of overfitting to SMOTE (high CV F1, low Test F1). Limitations include single-institution data and outcome ambiguity. Future improvements: Hyperparameter optimization, mutual information feature selection, and dashboard UI integration.